Executive Summary of Solar and Stellar Activity Cycle Forecasting with Gaussian Process Regression

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Exoplanetary research is a key avenue by which the origin of our planet and life on it can be studied. While over 6000 exoplanets have been discovered, the high-precision spectrographs required for exoplanet searches are oversubscribed: we cannot make all the measurements we desire. This project improves observation efficiency by using data from modest instruments to forecast the best times to observe stars with high-precision instruments, maximising exoplanet detections.

There are two main exoplanet detection methods: the transit method, which detects the dip in stellar luminosity caused by exoplanetary occlusion, and the radial velocity method, which detects the Doppler shift induced by an orbiting exoplanet. Non-exoplanetary noise in these measurements is produced by the formation of bright plages and dark spots on stellar surfaces.

The presence of these noise-generating phenomena is not constant. Their incidence varies cyclically, following stellar activity. For an Earth-like planet around a Sun-like star, observations at an activity minimum result in an order-of-magnitude improvement in detectability compared to observations at a maximum, which can be the difference between a detection and a null result.

If it were known when stars were going to be in their minima, observations could be prioritised accordingly to increase detection probabilities. Sairam & Triaud argued for publicly available stellar activity forecasting such that observations can be optimised. [1] This work set out to produce a scalable, automatic pipeline for this forecasting.

I. Key Achievements

The project's primary dataset is Mount Wilson Observatory S-Index data taken between 1966 and 1995; S-Index is a spectroscopic measure of stellar activity that can be calculated from modest instrumentation.

Conventional ARIMA and hybrid Fourier-ARIMA models were built first. These demonstrated the data's strongly cyclical nature and showed that fixed-form models are insufficiently flexible to capture the variations between cycles. This motivated the core focus of the work: the Gaussian process regression (GPR), which uses a kernel function to specify the covariances between the training data and the predictions.

GPRs were built around a physically motivated spectrum kernel: a sum of decaying oscillation modes representing the periodic covariances of stellar rotation, magnetic activity, and long-term trends in the S-Index data. The decay provides flexibility for the periodicities to evolve. The GPR computation was designed to scale linearly with dataset size to make it tractable for many stars.

A novel accomplishment was the creation of an automated algorithm to produce the optimisation priors for each kernel mode. This fulfilled a key requirement for the scalability of the pipeline by eliminating the need for manual prior tuning based on literature, which may not be available for all stars.

An iterative process using Lomb-Scargle periodograms was used to identify periodicities. These periodicities were checked for aliasing and observational window artefacts, and were then assigned to the physically motivated kernel modes based on population studies. The pipeline was built to handle cases where the data do not produce informative priors by falling back to population-level statistics reported by stellar surveys. When validated against benchmark stars with well-characterised rotation and stellar activity cycles, the rotation and activity cycle period priors found were consistent with the literature.

Not every target star exhibits all oscillation modes. To account for these physical scenarios, model comparison was performed. Priors for each mode were first found using the training data. Then, spectrum kernel GPR models were constructed with different combinations of modes to represent each distinct scenario. These were trained and the combination with the best validation performance was selected. By removing the need for the kernel modes to be manually tuned for each star, another requirement for pipeline scalability was fulfilled.

The chosen model's predictions were compared against test data using a realistic task of predicting the best time at which to observe stars in a given window. On the set of benchmark stars, forecast errors between predicted and actual best observation times for windows of up to five years remained below approximately one year, compared with the multi-year timescale of the variations. The result is a pipeline that is accurate on benchmark stars and designed to scale. Pending further verification by large-scale testing, this represents significant progress towards publicly available stellar cycle forecasting.

A further study to characterise how the cadence of observations affects forecast quality was performed with simulated stellar data, informing how future modest instrument observations should be taken to create the best stellar activity forecasts.

II. Next Steps

The project was limited by a prolonged HPC outage. As a result, a large-scale test run of the full set of active stars beyond the benchmarks could not be completed. The cadence analysis was also limited to 15 simulated stars rather than a larger sample, which would be more robust to noise fluctuations. These are a natural starting point for future work as they would more strongly validate the results. Following these validation tests, an up-to-date S-Index dataset should be compiled from spectroscopy data to produce operational forecasts from the pipeline.

Uncertainties are presently underestimated: the model accounts for uncertainty within the kernel fit, but not the uncertainty between kernel combinations. The methodology could be extended to use a reversible-jump MCMC to jointly quantify both types of uncertainty.

III. IMPACT

Oversubscribed high-precision instruments currently spend time observing stars regardless of where they are in their activity cycles. Consequently, many observations are taken of stars near their activity maxima, when exoplanets that could be detectable at activity minima are buried by stellar noise an order of magnitude larger than there. As such, the same telescope hours, scheduled around the activity minima, would yield more detections.

The design of the forecasting pipeline with automated prior search, automated kernel selection, and linear computational scaling with dataset size means that public forecasts of thousands of stars can be produced without needing per-star manual tuning or requiring costly observations.

The preliminary sampling cadence performance analysis further shows that the sampling rate a campaign needs depends on the forecast horizon desired. This informs the design of the observing programmes an operational forecast would rely on.

REFERENCES

[1] Lalitha Sairam and Amaury H M J Triaud. “The need for a public forecast of stellar activity to optimize exoplanet radial velocity detections and transmission spectroscopy”. en. In: *Mon. Not. R. Astron. Soc.* 514.2 (June 2022), pp. 2259–2268.